

Abhinav Tyagi

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PROJECTS

- **Pathos AI — Stateful Medical RAG Assistant** FastAPI, LangGraph, LangChain, Qdrant, FlashRank
AI/ML (Personal Project) 2026
 - **Agentic Orchestration:** Architected a clinical RAG assistant as a LangGraph state machine with conditional branching, routing unsafe requests (prompt injections, crisis language) away from the LLM and vector store entirely.
 - **Retrieval Pipeline:** Implemented a hybrid dense+sparse retrieval pipeline on Qdrant with FlashRank cross-encoder reranking and a hard token-budget constraint to improve context relevance over dense-only search.
 - **Safety & Privacy:** Designed a reversible PII tokenization layer and an output guardrail with bounded regeneration and safe-template fallback, eliminating raw PII exposure and definitive-diagnosis hallucinations.
- **Earthquake Prediction & Visualization App** Python, TensorFlow, scikit-learn, Pandas
AI/ML (Personal Project) 2025
 - **ML Pipeline:** Built an end-to-end pipeline — preprocessing, feature engineering, and Neural Network training — to predict earthquake magnitude and depth from historical seismic data.
 - **Performance:** Maintained ~92% prediction accuracy through hyperparameter tuning.
- **Online Food Ordering System** Java, DSA, Java Collections Framework
Backend & Cloud (Personal Project) 2026
 - **Architecture:** Developed a console-based food ordering system using Java and OOP principles to simulate real-world ordering functionality.
 - **Menu Management:** Implemented Array-List-based dynamic menu management for adding, removing, and updating food items.
 - **Order Processing:** Designed order placement and management logic using data structures to efficiently handle customer orders.
 - **OOP Implementation:** Applied encapsulation, inheritance, and polymorphism to create modular and maintainable system components.

EDUCATION

- **Vellore Institute of Technology (VIT-AP)** Andhra Pradesh, India
B.Tech. in Computer Science and Engineering (Core Specialization); GPA: 8.13 2023 – 2027 (Final-year)
- **Kendriya Vidyalaya** Chandigarh
Class XII (CBSE); 79% 2022
- **Kendriya Vidyalaya** Chandigarh
Class X (CBSE); 87.2% 2020

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, SQL, HTML5, CSS3
- **AI/ML & LLM Tooling:** LangChain, LangGraph, RAG, Prompt Engineering, TensorFlow, scikit-learn
- **Frameworks & Libraries:** React.js, Next.js, Express.js, Node.js, FastAPI
- **Databases & Infra:** MongoDB, MySQL, Docker, AWS
- **Core Concepts:** Data Structures & Algorithms, OOP, System Design Fundamentals, Operating Systems, Computer Networks

Achievement

- **Technical Team Lead — Data Analytics Club** Oct 2024 – Jul 2025
VIT-AP University Andhra Pradesh, India
 - **Community Leadership:** Led a 10-member technical team to run 6+ coding contests and workshops, growing an active developer community of 400+ students.
 - **Peer Learning:** Designed and launched a daily DSA challenge program to drive consistent peer learning and participation.

Languages

- English
- Hindi